

MAI-Transcribe-2-Streaming Model Card

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Overview

MAI-Transcribe-2-Streaming is a low-latency streaming speech recognition model built in-house by the Microsoft AI team that turns live audio into text as it is spoken. It continuously processes an audio stream and emits fast partial hypotheses followed by stable final results, so applications can respond while a person is still talking. This incremental output is well suited to live captions, voice agents, dictation, call experiences, and other interactive products where waiting for a complete recording would introduce unacceptable delay. It delivers reliable recognition across 60 languages and is designed to remain effective across varied accents, speaking styles, and real-world audio conditions.

Key capabilities

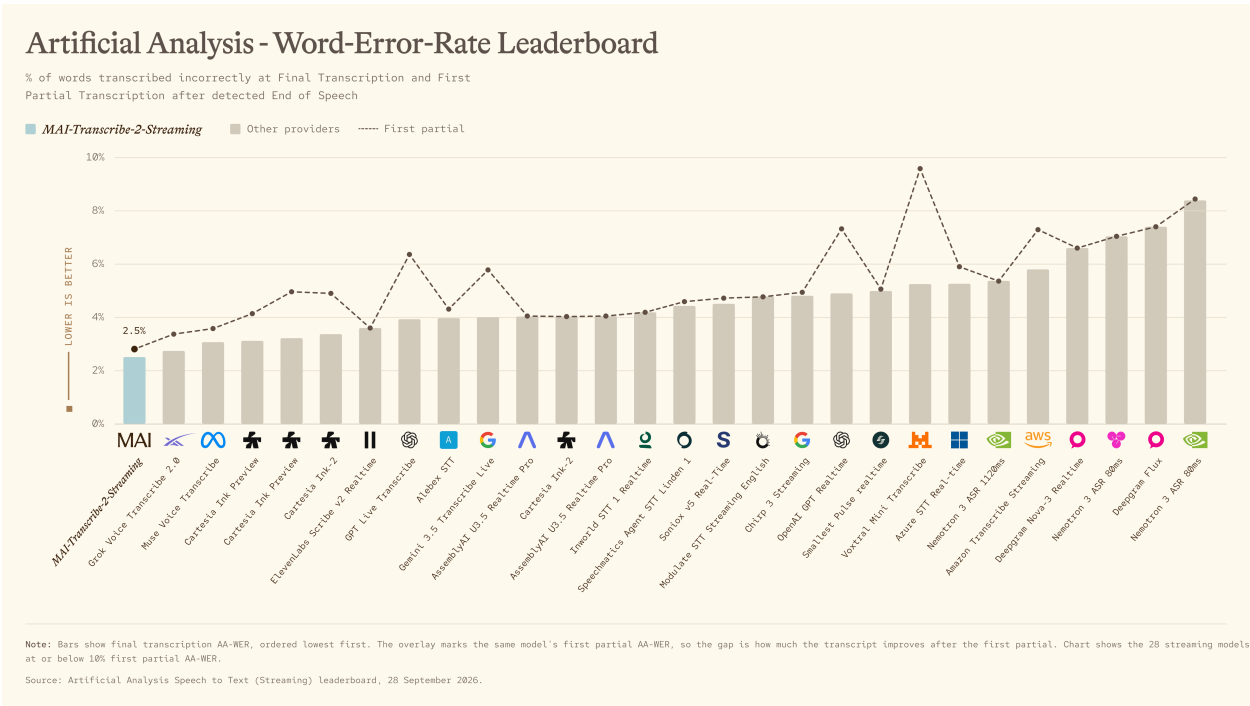
- Low-latency streaming recognition that processes audio continuously as it arrives.
- Immediate, evolving transcription for responsive user interfaces and downstream logic.
- Reliable speech recognition across 60 languages.
- Continuous session support for applications that need to recognize speech over an active audio connection.
- Robust recognition across varied accents, speaking styles, and real-world audio conditions.

Performance and quality

MAI-Transcribe-2-Streaming transcribes with high accuracy across all 60 supported languages, making it particularly interesting for global products. This is reflected by the Artificial Analysis benchmark where MAI-Transcribe-2-Streaming achieves the lowest percentage of words transcribed incorrectly at Final Transcription after detected End of Speech

Artificial Analysis - Word-Error-Rate Leaderboard to final transcription:

Model	Avg. Word Error Rate
MAI-Transcribe-2-Streaming	2.5%
Grok Transcribe 2.0	2.7%
Muse Voice Transcribe	3.1%
Cartesia Ink 2	3.1%
Eleven Labs Scribe V2 Realtime	3.6%
GPT-live Transcribe	3.9%
Alebex STT	4.0%



Use cases

Key use cases

- **Voice agents and assistants:** Stream microphone audio, use partial results to detect emerging intent, and act on finalized text once the utterance is complete.
- **Live captions and subtitles:** Display partial recognition results immediately, then replace them with stable final text as each utterance completes.



- **Contact center agent assistance:** Stream call audio and use finalized utterances to retrieve knowledge, populate forms, summarize context, and recommend next-best actions while the call is active.
- **Real-time call analytics:** Send finalized utterances to analytics and monitoring systems throughout the session to flag topics, intent, sentiment, compliance terms, or escalation risk.
- **Real-time dictation:** Render partial text as the user speaks and commit final results to the document with minimal perceived delay.
- **Meeting and collaboration experiences:** Process meeting audio continuously and make finalized utterances available to captions, notes, search, and summarization.
- **Interactive accessibility tools:** Convert live audio into continuously updated partial captions and stable final text across supported languages.
- **Live media and broadcasts:** Stream program audio to feed captions, moderation, live blogs, indexing, or translation workflows.
- **Live translation workflows:** Pass stable final utterances to downstream translation or speech synthesis as speakers complete each phrase.
- **Customer service automation:** Stream caller speech and route finalized requests to intent detection, workflow automation, or a human agent.

Pricing

\$0.54 per hour of audio through December 31, 2026.

Technical specs

Input formats

A continuous audio stream sent over an active connection. The model accepts audio as it arrives and transcribes it incrementally, without waiting for the recording to finish.

Output

Incremental text: fast partial hypotheses that update as more audio arrives, followed by a stable final transcript for each utterance.



Supported languages

60 languages:

Afrikaans, Arabic, Armenian, Assamese, Azerbaijani, Bengali, Bosnian, Bulgarian, Cantonese, Catalan, Chinese, Czech, Danish, Dutch, English, Estonian, Filipino, Finnish, French, Galician, German, Greek, Gujarati, Hebrew, Hindi, Hungarian, Icelandic, Indonesian, Italian, Japanese, Kannada, Kazakh, Korean, Latvian, Lithuanian, Macedonian, Malay, Malayalam, Marathi, Nepali, Norwegian Bokmål, Odia, Persian (Farsi), Polish, Portuguese, Punjabi, Romanian, Russian, Slovak, Slovenian, Spanish, Swahili, Swedish, Tamil, Telugu, Thai, Turkish, Ukrainian, Urdu, Vietnamese.

Distribution

Visit our API documentation to start integrating [MAI-Transcribe-API documentation](#)

You can call the model in one of two ways, depending on your preferred setup:

- OpenAI Realtime-compatible API over WebSockets: a drop-in option if you already build against the Realtime API, so existing integration patterns carry over with minimal changes.
- Azure Speech SDK: a drop-in option if you already build using the Azure Speech SDK.

MAI-Transcribe-2-Streaming is available in the following Azure regions:

- Central US
- East US 2 (Coming soon)
- Sweden Central
- Southeast Asia